DHOLE PATIL EDUCATION SOCIETY'S
DHOLE PATIL COLLEGE OF ENGINEERING
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ACA/R / 44	**Department of First Year Engineering**	Academic Year: 2025 -26
Rev : 00		Semester: ODD
Date: 15.12.2016	**CCE-II EXAM**	Section: A, B, C
		Subject- Basic Electrical Engineering
Duration : 1Hr		Max. Marks: 30

Instruction: **1) Assume suitable data, if necessary.**
2) Solve Q1 or Q2 and Q3 or Q4

Question Number		Question	Marks	BT	CO
Q.1		**QUESTION DISCRIPTION**			
	a	What is phase? Explain the concepts of phase lag and phase lead between two alternating quantities by i) Mathematical equations ii) Drawing waveforms and phasor diagrams	**04**	**3**	**CO3**
	b	The rms value of 50 Hz sinusoidal alternating current is 20A. At t=0, its value becomes 10A. Write down the equation for current. Also find the magnitude of current at t=6 ms.	**05**	**2**	**CO3**
	c	Define: i) cycle ii) time period iii) frequency of an alternating quantity iv)average value v) rms value vi) form factor & peak factor.	**06**	**1**	**CO3**
Q.2		**QUESTION DISCRIPTION**			
	a	Write the equation of instantaneous voltage in terms of: (i) Maximum voltage and angle θ (ii) Maximum voltage and angular velocity (iii) Maximum voltage and frequency (iv) Maximum voltage and time period. Draw sinusoidal waveform of voltage Vs time and show the maximum value, cycle, frequency and periodic time.	**04**	**2**	**CO3**
	b	A pure capacitance of 100uF is connected across single phase voltage given by v= 100 sin (314t) volts. Find; (i) Frequency of supply in Hz (ii) Capacitive reactance (in) Equation of curren	**05**	**3**	**CO3**
	c	A sinusoidally varying alternating voltage of 100 V (rms value) with 50 Hz frequency is applied to a circuit find : i) The mathematical equation of the voltage; ii) Time Period iii)The instantaneous voltage when t = 1.667 ms; iv) The time when instantaneous voltage is 100 V. v) Average value of the voltage vi) Maximum value of the voltage	**06**	**2**	**CO3**
Q.3		**QUESTION DISCRIPTION**			
	a	Derive the relation between i) phase voltage and line voltage ii) phase current and line current in case of balanced STAR connected 3-ph inductive load. Assume phase sequence RYB. Draw the circuit diagram & necessary phasor diagram.	**04**	**2**	**CO4**

	b	A series R-L-C circuit consists of R = 10 Ω, L = 0.318 H and C = 63.6 μF. This circuit is supplied by source of emf given by e(t) = 100 sin314t. Find: i) Expression for i (t) ii) Phase angle between voltage and current iii) Power factor of circuit iv) Active power consumed	**05**	**3**	**CO4**
	c	a)Define i) Balanced load ii) Unbalanced load iii) Symmetrical System b) Define active, reactive and apparent power in case of single phase circuit. State their units. Also draw the power triangle for R-L circuit.	**06**	**3**	**CO4**
Q.4		**QUESTION DISCRIPTION**			
	a	Derive the expression for power, when voltage v = Vm sin ωt is applied across R-L series circuit. Also draw the waveform for voltage, current & power on common X- axis.	**04**	**2**	**CO4**
	b	Three identical impedances each of (8 + j6) ohm are connected in star across 3-ph, 415 V, 50 Hz ac supply. Calculate: (i) line voltage, phase voltage, phase current, line current (ii) active power (iii) When same impedances are connected in delta across the same supply voltage, find active power.	**05**	**3**	**CO4**
	c	a) Define impedance. Draw impedance triangle for R-L and R-C series circuit. b) Derive the expression for resonant frequency.	**06**	**2**	**CO4**
		******* All the Best!*******			